

ABDUL RAHMAN

Backend Software Engineer

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PROFESSIONAL SUMMARY

Backend Engineer with 3+ years of experience building production-grade, cloud-native applications using Node.js, TypeScript, NestJS, and Express. Proven track record architecting scalable microservices, event-driven systems (Kafka, RabbitMQ), and multi-tenant SaaS platforms across AWS and Azure. Skilled in high-concurrency performance engineering — Redis caching strategies, single-flight cache-stampede protection, parallelized I/O, and TTL-based invalidation — alongside authentication/authorization (AWS Cognito, Keycloak), RESTful and real-time APIs (SSE, WebSockets, gRPC), and CI/CD automation. Experienced collaborating cross-functionally with Product, QA, and DevOps, and actively uses AI-assisted tools (Cursor, GitHub Copilot) to improve engineering productivity and code quality.

EXPERIENCE

Software Engineer | [Appinventiv Technologies Pvt. Ltd.](#) | June 2023 – Present

- Fixed slow page loads across portfolio, dashboard, and market-intelligence screens by architecting Redis-backed caching and pre-warming, and parallelizing independent I/O (valuations, anomaly-pipeline batches, version-key lookups) — turning sequential round-trips into concurrent execution.
- Eliminated cache stampedes and redundant compute under concurrent load by implementing single-flight cache-miss protection with wait/re-read coordination across portfolio, dashboard, and asset-detail handlers.
- Cut redundant Redis round-trips by up to 3× by parallelizing cache-version resolution (MGET/batch lookups), and designed short-TTL, publishing-safe caching for endpoints to replace slow full-period scans.
- Designed, developed, and maintained backend services and RESTful APIs using Node.js, TypeScript, NestJS, and Express across monolithic and microservices architectures, supporting millions of end users.
- Built cloud-native services on AWS (EC2, S3, CloudWatch, Lambda) and Azure, containerized with Docker, and implemented authentication/authorization with AWS Cognito and Keycloak (OAuth2/OIDC, RBAC) across multi-tenant platforms.
- Implemented MongoDB Change Streams to detect database updates and stream incremental changes via WebSockets to connected clients, enabling real-time UI updates without polling or manual refresh.
- Implemented event-driven workflows with Kafka and RabbitMQ and gRPC service-to-service calls, cutting inter-service latency ~20% vs. plain REST; built SSE-based real-time streaming for live client updates.
- Built and maintained CI/CD pipelines (GitHub Actions, GitLab CI), cutting release-related issues by ~40%, with 95%+ Jest test coverage.

KEY PROJECTS

QUIVIO (Sonnys) – Car Wash SaaS Platform [Link](#)

- Built an enterprise-level SaaS platform using microservices architecture in Node.js and TypeScript, serving 50+ enterprise clients with subscriptions, billing, and white-label configurations.
- Implemented authentication and authorization using AWS Cognito and Keycloak — including OAuth2/OIDC flows, role-based access control, and tenant-scoped identity management across 50+ enterprise tenants.
- Implemented Kafka-based event hubs for high-throughput event streaming and reliable asynchronous processing across services.
- Designed a multi-tenant PostgreSQL database architecture ensuring strict data isolation and scalability across tenants.
- Developed a dynamic user segmentation engine powering campaign execution with configurable scheduling and automated triggers.
- Implemented SendGrid email infrastructure with template management, inbound email parsing at sub-user level, and sender identity verification for secure, scalable communication.

AMERICAN EXPRESS – Chargeback Portal [Link](#)

- Developed a microservices-based, multi-layered backend architecture for a chargeback management platform serving multiple merchant banks.
- Enabled inter-service communication using gRPC for synchronous flows and RabbitMQ for asynchronous processing across layers and services.

- Designed and maintained tenant-isolated PostgreSQL databases, ensuring data consistency, reliability, and scalability for finance-grade use cases.

BARAKAH – Food & Restaurant Platform [Link](#)

- Built a monolithic Node.js backend for an online food ordering and restaurant management system, handling order lifecycle, payments, notifications, and real-time order tracking via WebSockets.
- Designed REST APIs supporting high concurrency and peak traffic for 2M+ users; optimized MongoDB queries for trending meals, popular restaurants, and personalized recommendations.
- Used Redis for caching frequently accessed data, reducing database load and improving response times for high-traffic endpoints.
- Leveraged MongoDB Change Streams to push near real-time updates to clients, eliminating manual page refreshes and improving user experience via WebSockets.

YKA CONNECT – Automobile Services Platform [Link](#)

- Developed backend services for booking, leasing, servicing, test drives, and payments using Node.js and MongoDB, with geo-location-based discovery.
- Designed and implemented a global search engine serving 5K+ active users, and a multi-tier loyalty and rewards system with tier upgrades and activity-based progression rules.
- Built a multi-channel notification system delivering push, email, and SMS notifications concurrently to thousands of users, ensuring reliable, real-time delivery at scale.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript

Backend: Node.js, NestJS, Express, Hapi, Koa

APIs & Real-time: RESTful APIs, gRPC, WebSockets, SSE (Server-Sent Events), GraphQL (familiar)

Caching: Redis (TTL strategies, cache pre-warming, single-flight/mutex concurrency control, MGET batch lookups, cache invalidation)

Auth & Identity: AWS Cognito, Keycloak, OAuth2/OIDC, JWT, RBAC

Architecture: Microservices, Event-Driven Systems, Multi-Tenant Design, Cloud-Native

Databases: MongoDB, PostgreSQL, MySQL

Messaging: Kafka, RabbitMQ

Cloud & DevOps: AWS (EC2, S3, Lambda, CloudWatch), Azure, Azure Key Vault, APIM, Docker, GitHub Actions, GitLab CI

Monitoring & Logging: Datadog, Grafana, AWS CloudWatch, Sentry, Prometheus

AI-Assisted Dev: Cursor, GitHub Copilot, ChatGPT

Testing: Jest (Unit & Integration, 95%+ coverage)

Integrations: Stripe, PayFort, TAP, Twilio, SendGrid

EDUCATION

B.Tech – Computer Science & Engineering

Sanskar College of Engineering & Technology, Ghaziabad

2019 – 2023

CERTIFICATIONS

- [AWS Cloud Practitioner Essentials](#) — Amazon Web Services